

REPLICATION EXERCISE: OF THE EXPERIMENTAL ESTIMATES OF EDUCATION

FUNCTIONS

BY

ALAN B. KRUEGER

OLADINI AYOMIDE BLESSING

UNIVERSITY OF NORTHERN BRITISH COLUMBIA

ECON 712

230168183

Table of Contents

1. Introduction and Contribution	3
2.1 The identification problem	5
2.2 Strategy	6
2.3 Empirical specification	8
2.4 Critical assessment.....	10
3. Replication.....	11
3.1 Data.....	11
3.2 Descriptive Statistics	13
3.3 Main Results	15
4. Critical Discussion and Proposed Extensions.....	17
4.1 Assessment of Empirical Choices	17
4.2 Proposed Improvements and Extensions	18
4.3 Overall Assessment	18
5. Conclusion	19

1. Introduction and Contribution

This paper replicates the empirical analysis of Alan Krueger (1999), which studies whether reducing class size improves student achievement using data from the Tennessee STAR experiment. The main question is simple but very important: do smaller classes actually lead to better academic outcomes, especially in the early years of schooling?

This question matters because class size is one of the most debated education policies, and it is also very expensive to implement. If smaller classes really improve student performance, then the cost might be justified. But if the effect is small or unclear, then those resources could be used elsewhere. The difficulty is that most of the earlier literature does not give a clear answer. A big reason for that is that class size is usually not randomly assigned. For example, weaker students might be placed in smaller classes, or better schools might have smaller classes. So when we see that smaller classes perform better, it is not obvious whether that is a causal effect or just selection.

This is where Krueger (1999) stands out. The paper uses the STAR experiment, where students and teachers were randomly assigned to small classes, regular classes, or regular classes with a teacher's aide within schools. That random assignment is a big deal because it removes the usual selection problem. It means that, on average, students in different class types should look similar before the treatment, so any differences in outcomes can be interpreted as coming from class size.

That said, the experiment is not perfectly clean. Students move between classes, some drop out of the sample, and we do not observe baseline test scores before assignment. So even though the

design is strong, there are still some practical issues. This is why the paper, and my replication, also rely on additional methods like instrumental variables to deal with these imperfections.

The main findings of the original paper are that students in small classes perform better on standardized tests, especially in kindergarten and first grade, and that these gains tend to persist over time. In contrast, having a teacher's aide in a regular-sized class does not seem to make much difference. Because of the randomized design, the paper argues that these effects are causal.

My replication confirms these results. Across all entries, students assigned to small classes are in clearly smaller classrooms, with average class sizes around 15–16 students compared to over 22 students in regular classes. At the same time, they consistently achieve higher test scores. For example, in first grade, the gap in performance is quite noticeable, and the regression results show gains of up to about 8.6 percentile points in some specifications. Even after adding controls and school fixed effects, the estimated effects remain positive and statistically significant, generally in the range of about 3 to 5 percentile points in later grades. (Krueger, 1999, pp. 497-532)

Once we move to instrumental variable estimates, the results become even clearer. Using initial class assignment as an instrument for actual class size, I find that increasing class size reduces test scores, and the effect is larger than what we get from OLS. For example, in first grade, the IV estimate suggests that adding one more student to a class reduces test scores by about 0.85 percentile points. This tells us that OLS is likely understating the true effect, which makes sense if class size is not perfectly random in practice. Overall, the replication supports the main conclusion of the original study: smaller classes lead to better student outcomes, especially in the early grades. The results are consistent across different specifications and group, which makes the causal interpretation quite convincing.

2. Identification Strategy

The main reason this paper matters so much is that it tries to answer a causal question, not just a descriptive one. The question is not simply whether students in smaller classes do better. It is whether being placed in a smaller class actually causes better academic performance. This distinction is important, because in most real-world settings class size is not randomly determined. Schools may place weaker students into smaller classes, parents may push for certain classroom placements, and better-funded schools may be able to maintain lower student-teacher ratios. So if we just compared outcomes across small and large classes without thinking carefully about identification, we could easily mistake selection for causation.

2.1 The identification problem

The fundamental identification challenge in this paper is that class size is usually endogenous. In a naive comparison, students in smaller classes may differ from students in larger classes in ways that also affect achievement. For example, if lower-performing students are intentionally placed in smaller classes for extra support, then a simple comparison might understate the true benefit of smaller classes. Class size would be correlated with omitted factors such as prior ability, parental involvement, school resources, or teacher quality. That makes it hard to know whether differences in test scores are really caused by class size or by these other factors.

This is exactly the problem Krueger is trying to solve. Krueger's contribution is to bring in an experimental design that breaks that link and allows class size to be treated, at least initially, as exogenous.

2.2 Strategy

The main identification strategy is the randomized design of Project STAR. Students and teachers were randomly assigned within schools to one of three class types: small classes, regular classes, and regular classes with a teacher's aide. Small classes were intended to have about 13 to 17 students, while regular classes were around 22 to 25 students. Because assignment took place within schools, the idea is that students in different treatment groups should be comparable at baseline, except for the classroom environment to which they were assigned.

This matters because if randomization is successful, then assignment to class type is unrelated to unobserved student characteristics. In that case, differences in later test scores can be interpreted as causal effects of class size rather than simple correlations. In the paper, Krueger supports this by showing that observable background characteristics are generally balanced across groups within schools. My replication reaches the same broad conclusion. In Table 1, students assigned to small classes are consistently placed in much smaller classrooms, but background characteristics such as free lunch status and age are broadly similar across treatment groups.

Still, the paper does not stop at random assignment alone, because the experiment was not implemented perfectly. Krueger explains several complications. First, students in regular classes were rerandomized between regular and regular/aide classes after kindergarten. Second, some students moved between class types over time. Third, there was substantial attrition, with many students missing in later grades. Finally, actual class size did not always match the intended treatment because of parents' mobility and enrollment changes. These issues mean that actual class size is not perfectly randomized, even though initial assignment is.

That is why the paper also uses instrumental variables. The instrument is initial class assignment, and it is used to identify the causal effect of actual class size on test scores. Initial assignment is determined by the experiment, so it is plausibly exogenous. At the same time, it strongly predicts actual class size, because students assigned to small classes actually sit in substantially smaller classrooms. My replication shows this clearly. In Table 3, average class size is about 15.68 in small classes, compared with 22.67 in regular classes and 23.40 in aide classes. (Krueger, 1997) That strong first-stage relationship is what makes the IV strategy credible.

For this strategy to deliver causal estimates, several assumptions are needed. The first is that it is a random assignment, meaning initial assignment is unrelated to unobserved determinants of test scores. The second is that the initial class assignment must strongly predict actual class size. The third is the exclusion restriction, meaning assignment affects achievement only through class size and not through some other channel. Also, stable unit treatment is assumed, meaning one student's class assignment does not directly affect another student's outcome across classrooms in a way that would bias estimation.

Overall, these assumptions are reasonably plausible in the context of Project STAR. Krueger was careful about these issues, which is one reason the paper is stronger than a simple experimental paper. He does not just assume the experiment worked, He treats the imperfections as part of the empirical problem and adjusts the analysis accordingly.

2.3 Empirical specification

The main reduced-form specification estimated in the paper regresses student test scores on indicators for class type, along with controls and school fixed effects. In simplified form, the model can be written as:

$$Score_{ics} = \beta_0 + \beta_1 Small_{cs} + \beta_2 RegAide_{cs} + X'_{ics}\gamma + \delta_s + \varepsilon_{ics}$$

where $Score_{ics}$ is the test score of student i in class c and school s , $Small_{cs}$ is a dummy variable equal to one if the student is assigned to a small class, and $RegAide_{cs}$ is a dummy for assignment to a regular-sized class with a teacher's aide. The vector X_{ics} includes observed student and teacher characteristics, such as race, gender, free lunch status, teacher experience, and teacher education, while δ_s represents school fixed effects. The constant term β_0 captures the average test score for students in the omitted category, which is a regular-sized class without an aide, holding other variables constant. In this specification, β_1 measures the effect of being in a small class relative to a regular class, while β_2 captures the effect of being in a regular class with an aide.

This is the specification I replicate in Table 5. Across grades, the coefficient on the small class dummy is positive and statistically significant in nearly every specification. In kindergarten, the estimated effect is about 5.3 percentile points once school fixed effects are included. In first grade, the effect is even larger, reaching about 8.3 points, remaining above 5 points even with additional controls. In second and third grade, the estimates are smaller but still positive, generally around 3 to 6 points depending on the specification. By contrast, the coefficient on the aide dummy is usually small and often statistically insignificant, which matches the original paper's conclusion that aides are not a substitute for smaller classes.

The paper also estimates the effect of actual class size directly using OLS and 2SLS. In that case, the basic model becomes:

First stage

$$ClassSize_{ics} = \pi_0 + \pi_1 Small_{cs} + \pi_2 RegAide_{cs} + X'_{ics}\gamma + \delta_s + u_{ics}$$

Second stage

$$Score_{ics} = \beta_0 + \theta ClassSize_{ics} + X'_{ics}\gamma + \delta_s + \varepsilon_{ics}$$

where $ClassSize_{ics}$ is the actual number of students in the classroom for student i in class c and school s , and $Small_{cs}$ and $RegAide_{cs}$ are dummy variables indicating the student's initial class assignment. These variables serve as instruments for actual class size. The vector X_{ics} includes student and teacher characteristics, and δ_s represents school fixed effects.

In the first stage, class size is regressed on initial assignment and controls, which isolates the variation in class size that comes from random assignment. In the second stage, test scores are regressed on the predicted class size from the first stage. The coefficient θ captures the causal effect of class size on student achievement. Krueger instruments for it using the initial assignment to class type. That two-stage strategy is what I replicate in Table 7.

The controls included in these models are there to improve precision and absorb remaining observable differences across students and teachers. These include race, gender, free lunch status, teacher race, teacher gender, teacher experience, and teacher education, along with school fixed effects. At the same time, one important omitted variable remains baseline achievement, because the dataset does not contain pre-treatment test scores. That is a real limitation, since baseline test scores would have provided a very direct check of whether treated and control students were similar before assignment.

2.4 Critical assessment

In my view, the identification strategy is convincing overall, and this is one of the main strengths of the paper. The randomized design solves the biggest problem that usually affects class size studies, and the use of school fixed effects makes sense because randomization occurred within schools rather than across the full sample. The instrumental variables strategy is also well motivated, since actual class size is not perfectly random after treatment assignment. Taken together, these methods give the paper a strong claim to causality.

However, one concern is attrition. Krueger notes that a large share of students leave the sample over time, and if attrition is related to treatment status or student potential outcomes, estimates may still be biased. In my replication, attrition is statistically different across groups for kindergarten entrants, which suggests this issue is not just theoretical. Another concern is class switching. Once students move across class types, treatment exposure no longer lines up perfectly with initial randomization. IV helps address this, but it also means the estimates are identifying a local average treatment effect for students whose actual class size responds to assignment, not necessarily the effect for every student.

Even with these, I still think the paper does a strong job of establishing causality. The threats to identification are there, but they are mostly handled carefully. More importantly, the main results are consistent across descriptive evidence, reduced-form regressions, and IV estimates. That consistency makes the overall conclusion hard to dismiss that is smaller classes improve student achievement, especially in the early grades.

3. Replication

3.1 Data

The data used in this paper come from Project STAR, the Tennessee Student/Teacher Achievement Ratio experiment. This is the same dataset used by Krueger (1999) to study the effect of class size on student achievement. Project STAR followed students and teachers in Tennessee public schools from kindergarten through third grade and randomly assigned students to one of three class types: small classes, regular classes, and regular classes with a teacher's aide. The experiment was designed to create exogenous variation in class size, which is what makes it so useful for causal analysis.

According to the assignment instructions, the STAR data are publicly available through Harvard Dataverse, and that is the source used for this replication. The original paper works with student-level data linked to classroom and teacher information, so the replication also required combining information across these dimensions in order to reproduce the main tables and regressions.

To carry out the replication, I first obtained the STAR dataset and identified the variables needed for the tables and regressions up to page 519 of the original paper, as required by the assignment. I then cleaned the data by identifying variable names, constructing class-type indicators, and creating outcome measures and control variables consistent with the published analysis. In particular, I generated indicators for assignment to small classes and regular class, and regular classes with an aide, as well as controls such as race, gender, free lunch status, teacher race, teacher gender, teacher experience, and teacher education. Because the original paper estimates models separately by grade, I also organized the data by entry and school year.

One of the main challenges in the cleaning process was that the available dataset did not line up perfectly with the published tables. For instance, the construction of the test score variable. The

original paper uses average percentile scores based on Stanford Achievement Test components, while in my earlier replication work, I initially used a raw average of test components. That produced the same broad pattern of results, but the levels were not directly comparable to the paper. I later reconstructed the score variable using percentile-based components, which made my results much closer to the original table values. Another issue was race. The published paper reports a White/Asian measure, but the version of the dataset available to me required approximating this variable using the available coding. This likely explains some of the remaining differences between my balance tables and the published ones.

Attrition and sample definition were also important challenges. Because students enter and leave the sample over time, not every observation can be used in every table. For example, attrition is not available for students who enter in third grade because there is no later grade in the data to observe whether they remain in the sample. Similarly, reproducing transition tables required restricting the data to students observed in consecutive grades, which reduced the usable sample relative to the full dataset. In general, when exact replication was not possible because of variable construction or sample restrictions, I kept the definitions as close as possible to the original study and explained the differences clearly in the discussion of results.

Overall, although there were some practical data limitations, the dataset is rich enough to replicate the main descriptive tables, figure, reduced-form regressions, and IV estimates from the paper. The final cleaned data produce results that are broadly consistent with Krueger (1999), especially with respect to the direction, significance, and overall magnitude of the class size effects.

3.2 Descriptive Statistics

In this paper, the main descriptive tables are Table 1, Table 2, Table 3, and Table 4, together with Figure I. These results show very clearly; 1. The treatment was implemented successfully, and 2. students in small classes generally performed better than students in regular classes.

Table 1 compares students assigned to small classes, regular classes, and regular classes with an aide across the different entry cohorts. The main purpose of this table is to check whether random assignment produced groups that look similar at baseline while also confirming that the treatment generated large differences in actual class size. My replication shows that this is broadly the case. Across all four entry cohorts, students assigned to small classes are placed in substantially smaller classrooms, with average class sizes around 15 to 16 students, compared with roughly 22 to 24 students in regular classes. At the same time, students in small classes tend to have higher average test scores. For example, among kindergarten entrants, the average percentile score is about 51.8 in small classes, compared with 48.04 in regular classes and 50.2 in regular classes with an aide. A similar pattern appears in first grade, where the gap is even larger: 53.6 in small classes versus 46.55 in regular classes.

Table 2 strengthens this point by reporting joint tests of equality across treatment groups after controlling for school fixed effects. This matters because assignment in STAR was random within schools, not across the full statewide sample. In my replication, most background characteristics are not significantly different across groups once school effects are included, especially for free lunch status and age. The score variable is also significant in most cohorts, which is consistent with the paper's main claim that smaller classes improve performance.

Table 3 focuses on the distribution of actual class size in first grade. This table is useful because it moves beyond average differences and shows whether the treatment groups are really separated in

practice. My results show a very clear pattern. Students assigned to small classes are concentrated in classrooms with roughly 13 to 17 students, while those assigned to regular or aide classes are concentrated in the low to mid-20s. The average class size in my replication is 15.68 for small classes, 22.67 for regular classes, and 23.40 for aide classes, which is very close to the original paper. This strong separation is important because it also supports the relevance of initial assignment as an instrument in the IV analysis.

Table 4 examines transitions between class types across grades. This matters because if students move across treatment groups, actual treatment exposure may not match initial assignment. My replication shows that a large share of students remains in the same class type, especially students initially assigned to small classes. At the same time, there is noticeable movement between regular and regular-plus-aide classes, and some movement across treatment groups more generally. This is one of the reasons the paper does not rely only on comparisons based on actual class size and instead also uses instrumental variables based on initial assignment.

Finally, Figure I compares the distribution of student achievement across class types. In my replication, the score distribution for students in small classes is generally shifted to the right relative to the distribution for regular classes, especially in kindergarten and first grade. This suggests that the gains from smaller classes are not driven by only a few extreme observations but reflect a broader improvement in performance across the distribution.

Overall, the descriptive evidence lines up well with the original paper. The treatment groups are clearly different in class size, broadly similar in most observable characteristics, and students in small classes tend to perform better on average. That gives a strong foundation for the regression analysis that follows.

3.3 Main Results

The main regression results are reported in Table 5, Table 6, and Table 7. These are the core results of the replication because they show whether the positive relationship between small classes and achievement remains after controls are added and whether the effect can be interpreted as causal. Table 5 reproduces the main reduced-form regressions of student test scores on class-type indicators. The key coefficient is the one on the small class dummy, which measures the effect of being assigned to a small class relative to a regular class without an aide. My replication strongly supports the original paper's findings. In kindergarten, the estimated effect of a small class is about 5.3 percentile points once school fixed effects are included. In first grade, the effect is even larger, reaching about 8.6 percentile points. In second and third grade, the coefficients are somewhat smaller but still positive and statistically significant, generally in the range of about 3 to 6 percentile points depending on the specification. This pattern is very much in line with the original paper: the effect of smaller classes is strongest in the early grades and remains positive through the year.

By contrast, the regular-with-aide treatment does not show the same consistent benefit. In my results, the coefficient on the aide dummy is often small and statistically insignificant, especially once controls are added. This is also consistent with Krueger's conclusion that adding a teacher's aide to a regular-sized class is not an effective substitute for reducing class size itself.

Table 6 compares estimates based on actual class assignment with those based on initial assignment. This comparison matters because some students switch class types over time, so actual class type may partly reflect post-randomization behavior rather than pure treatment assignment. In my replication, the estimates based on initial assignment are generally slightly smaller than those based on actual class assignment, but they remain clearly positive and statistically significant.

This is exactly what we would expect if switching weakens treatment intensity. It also supports the paper's argument of the main findings

Table 7 is especially important because it estimates the effect of actual class size directly using both OLS and two-stage least squares. The OLS results show a negative relationship between class size and achievement in every grade. In my replication, one additional student reduces test scores by about 0.63 percentile points in kindergarten, 0.82 in first grade, 0.42 in second grade, and 0.36 in third grade. However, the IV estimates are even more negative: about -0.71 in kindergarten, -0.85 in first grade, -0.49 in second grade, and -0.65 in third grade. This pattern suggests that OLS understates the true causal effect of class size.

Overall, the replication was successful, although my results are not exactly identical to the published paper in every number, but they are consistent. The descriptive statistics, regression tables, and IV estimates all support the same conclusion as the original paper: smaller classes improve student achievement, especially in the early grades, while teacher aides do not generate comparable gains. The results are robust across multiple specifications, and the remaining differences from the published paper do not change the basic interpretation.

4. Critical Discussion and Proposed Extensions

4.1 Assessment of Empirical Choices

One of the main strengths of Krueger (1999) is that the empirical design is much stronger than what is common in the class size literature. Instead of relying on observational data, the paper uses Project STAR, a randomized experiment in which students and teachers are assigned within schools to small classes, regular classes, or regular classes with a teacher's aide. That choice already solves a major part of the identification problem.

Starting out, I think the paper makes a sensible choice by focusing on standardized test scores. Since the question is whether smaller classes improve student learning in the short run, test scores are a natural outcome to study. They are measured consistently across students and grades and give a direct way to compare achievement. However, they captured only one dimension of learning and may miss outcomes such as motivation, persistence, later educational attainment, or labor market success. So, while I agree with the paper's choice for the main analysis, it narrows the interpretation of the results.

On the control variables, the models include student characteristics such as race, gender, and free lunch status, along with teacher characteristics and school fixed effects. In principle, randomization means these controls are not needed to identify the treatment effect. But in practice they can improve precision and absorb small imbalances that arise in any actual dataset.

On the functional form, the main models are linear regressions, either in reduced form using class-type dummies or in IV form using actual class size as a continuous explanatory variable. It made the coefficients easy to interpret, and the main question is naturally linear: how much do scores change when students move from a regular class to a smaller class, or when class size increases by one student? A log-linear model would be harder to interpret here.

4.2 Proposed Improvements and Extensions

An improvement I would make is to connect early class size exposure to longer-run outcomes. The original paper focuses on test scores in kindergarten through third grade, which is reasonable given the available data in the analysis, but that also leaves an important question unanswered. 1. do these early gains matter in the long run? Higher test scores in the short run are important, but policymakers ultimately care about whether smaller classes translate into outcomes such as later achievement, high school completion, college attendance, and labour market performance. A follow-up paper that linked the STAR data to later educational records would provide a much broader view of the returns to class-size reduction.

This extension would require access to longitudinal administrative data or linked follow-up surveys. The empirical strategy could remain close to the original design by exploiting the same random assignment. The advantage of this extension is that it would move the discussion from “do smaller classes raise test scores in early grades?” to “do smaller classes have lasting economic value?” That would be especially useful in evaluating whether the policy is worth its cost.

4.3 Overall Assessment

Overall, I do find the paper’s conclusions convincing. I liked the combination of random assignment, reduced-form estimates, and instrumental variables gives the analysis a level of credibility that most class-size studies do not have. My own replication reinforces that view.

If I were approaching the research question myself, I would make is to address attrition more explicitly. While the original paper acknowledges that a significant number of students leave the sample over time, it does not fully explore whether attrition is systematically related to treatment status or potential outcomes. This matters because if students who leave the sample differ in achievement in a way that is correlated with class assignment, the estimated effects may be biased.

5. Conclusion

This paper replicates the main empirical findings of Alan Krueger (1999) using data from the STAR experiment, with the aim of assessing whether smaller class sizes improve student achievement and whether this relationship can be interpreted as causal. Overall, the replication is successful and supports the central conclusions of the original study.

The results are consistent. Students assigned to small classes are placed in significantly smaller classrooms and achieve higher test scores. The effects are strongest in the early grades, particularly in first grade, and remain positive in later grades. In contrast, the results show that assigning a teacher's aide to a regular-sized class does not generate similar improvements.

The instrumental variables result further strengthen these findings. By using initial class assignment as an instrument for actual class size, the analysis isolates the variation in class size that comes from random assignment. The fact that IV estimates are more negative than OLS suggests that OLS understates the true effect, confirming that the relationship between class size and achievement is causal rather than driven by selection.

At the same time, the analysis highlights some limitations. Issues such as attrition, class switching, and the absence of baseline test scores.

Overall, the evidence is clear and consistent: smaller classes improve student achievement, especially in the early years of education. The results are economically meaningful but not extremely large, which implies that policy decisions should weigh these benefits against the cost of reducing class size. This replication confirms the strength of the original paper while also pointing to areas for future research.

References

KRUEGER, A. B. (1999).

Experimental estimates of education production functions. *Quarterly Journal of Economics*, 114(2), 497–532.

KRUEGER, A. B., & WHITMORE, D. M. (2001).

The effect of attending a small class in the early grades on college-test taking and middle school test results: Evidence from Project STAR. *Economic Journal*, 111(468), 1–28.

TENNESSEE STAR PROJECT DATA.

Project STAR public-use data set. Harvard Dataverse.

TABLES

TABLE 1

COMPARISON OF MEAN CHARACTERISTICS OF TREATMENTS AND CONTROLS

A. Students Who Entered STAR in Kindergarten

Variable	Small Class	Regular Class	Regular + Aide
Free Lunch	0.52	0.52	0.49
White/ Asian	0.86	0.80	0.84
Age in 1985	5.77	5.75	5.75
Class Size	15.12	22.38	22.8
Percentile Score	52.15	47.4	47.2
Attrition Rate	0.277	0.32	0.30

B. Students Who Entered STAR in First Grade

Variable	Small Class	Regular Class	Regular + Aide
Free Lunch	0.57	0.61	0.60
White/Asian	0.88	0.78	0.77
Age in 1985	6.00	6.14	6.17
Class Size	15.83	22.69	23.41
Percentile Score	53.6	46.55	50.31
Attrition Rate	0.41	0.39	0.38

C. Students Who Entered STAR in Second Grade

Variable	Small Class	Regular Class	Regular + Aide
Free Lunch	0.66	0.63	0.65
White/Asian	0.88	0.78	0.83
Age in 1985	6.17	6.19	6.22
Class Size	15.48	23.7	23.56
Percentile Score	53.07	50.28	46.55
Attrition Rate	0.397	0.382	0.383

D. Students Who Entered STAR in Third Grade

Variable	Small Class	Regular Class	Regular + Aide
Free Lunch	0.60	0.63	0.69
White/ Asian	0.82	0.73	0.63
Age in 1985	6.24	6.22	6.27
Class Size	15.91	24.07	24.34
Percentile Score	53.06	50.17	46.33
Attrition Rate	NA	NA	NA

Notes: Values are group means by class type. Test scores are reconstructed as percentile-based averages of the math, reading, and word SAT components, Race is approximated using the available dataset. Attrition is measured using missing next-grade math scores and is not available for third-grade entrants.

TABLE 2
P-VALUES FOR TESTS OF WITHIN-SCHOOL DIFFERENCES AMONG SMALL,
REGULAR,
AND REGULAR/AIDE CLASSES

Variable	Kindergarten	First Grade	Second Grade	Third Grade
Free Lunch	0.56	0.34	0.62	0.17
White/Asian	0.69	0.30	0.17	0.27
Age in 1985	0.40	0.12	0.48	0.39
Attrition Rate	0.00	0.10	0.50	N/A
Actual Class Size	0.00	0.00	0.00	0.00
Percentile Score	0.00	0.00	0.53	0.00

Notes: Entries are p-values from joint F-tests of equality across small, regular, and regular-plus-aide classes, conditional on school fixed effects. Test scores are constructed as percentile averages of math, reading, and word assessments. Attrition is defined based on missing next-grade test scores and is not available for third-grade entrants.

TABLE 3
DISTRIBUTION OF CHILDREN ACROSS ACTUAL CLASS SIZES BY RANDOM
ASSIGNMENT
GROUP IN FIRST GRADE

Assignment group in first grade			
Actual class size in first grade	Small	Regular	Aide
12	24	0	0
13	182	0	0
14	252	0	0
15	465	0	0
16	256	16	0
17	561	17	0
18	108	36	0
19	57	76	57
20	20	200	120
21	0	378	378
22	0	594	330
23	0	437	460
24	0	384	264
25	0	175	225

26	0	130	234
27	0	54	108
28	0	28	56
29	0	29	58
30	0	30	30
Average class size	15.68	22.67	23.40

Note: Actual class was determined by counting the number of students in the data set with the same class identification.

TABLE 4
TRANSITIONS BETWEEN CLASS-SIZE IN ADJACENT GRADES
NUMBER OF STUDENTS IN EACH TYPE OF CLASS

A. Kindergarten to first grade				
Kindergarten	Small	Regular	Reg/Aide	All
Small	1203	56	44	1303
Regular	119	688	618	1425
Aide	111	713	666	1490
All	1433	1457	1328	4218
B. First grade to second grade				
First Grade	Small	Regular	Reg/Aide	All
Small	1195	12	18	1225
Regular	124	1187	138	1449
Aide	23	84	1307	1414
All	1342	1283	1463	4088
C. Second grade to third grade				
Second Grade	Small	Regular	Reg/Aide	All
Small	1291	28	23	1342
Regular	126	1063	94	1283
Aide	27	43	1393	1463
All	1444	1134	1510	4088

TABLE 5
OLS AND REDUCED-FORM ESTIMATES OF EFFECT OF CLASS-SIZE ASSIGNMENT
ON
AVERAGE PERCENTILE OF STANFORD ACHIEVEMENT TEST

OLS: actual class size				Reduced form: initial class size				
A. Kindergarten								
Explanatory Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)

Small class	4.73	5.32	5.35	5.35	4.73	5.32	5.35	5.35
	(2.18)	(1.26)	(1.21)	(1.19)	(2.18)	(1.26)	(1.21)	(1.19)
Regular/aide class	-0.01	0.15	0.41	0.16	-0.01	0.15	0.41	0.16
	(2.21)	(1.13)	(1.09)	(1.07)	(2.21)	(1.13)	(1.09)	(1.07)
White/Asian	—	—	-1.17	-1.03	—	—	-1.17	-1.03
			(2.16)	(2.15)			(2.16)	(2.15)
Girl	—	—	4.46	4.38	—	—	4.46	4.38
			(0.63)	(0.63)			(0.63)	(0.63)
Free lunch	—	—	-14	-13.94	—	—	-14	-13.94
			(0.76)	(0.76)			(0.76)	(0.76)
White teacher	—	—	—	-0.59	—	—	—	-0.59
				(2.12)				(2.12)
Teacher experience	—	—	—	0.25	—	—	—	0.25
				(0.1)				(0.1)
Master's degree	—	—	—	-0.52	—	—	—	-0.52
				(1.06)				(1.06)
School FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
R ²	0.01	0.25	0.30	0.30	0.01	0.25	0.30	0.30

B. First grade

Explanatory Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small class	8.6	8.31	7.19	6.73	7.46	6.95	5.76	5.49
	(1.9)	(1.15)	(1.19)	(1.22)	(1.7)	(1.09)	(1.13)	(1.14)
Regular/aide class	3.38	2.17	1.59	1.12	1.83	1.54	0.34	0.3
	(1.96)	(0.95)	(1.15)	(1.17)	(1.07)	(0.77)	(0.82)	(0.82)
White/Asian	—	—	-0.6	-0.18	—	—	-0.61	-0.17
			(1.67)	(1.64)			(1.71)	(1.66)
Girl	—	—	4.53	4.56	—	—	4.44	4.49
			(0.65)	(0.65)			(0.65)	(0.65)
Free lunch	—	—	-14.45	-14.57	—	—	-14.53	-14.66
			(1.04)	(1.04)			(1.05)	(1.05)
White teacher	—	—	—	-3.95	—	—	—	-4.08
				(2.14)				(2.16)
Male teacher	—	—	—	6.46	—	—	—	7.7
				(3.23)				(3.02)
Teacher experience	—	—	—	0.04	—	—	—	0.04
				(0.06)				(0.06)

Master's degree	—	—	—	2.68	—	—	—	3.45
				(1.56)				(1.4)
School FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes

C. Second grade

Explanatory Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small class	5.59	6.08	4.07	4.1	5.24	5.52	3.21	3.18
	(1.9)	(1.23)	(1.35)	(1.35)	(1.62)	(1.1)	(1.16)	(1.17)
Regular/aide class	1.08	1.38	1.04	0.99	-0.08	1.06	-0.12	-0.09
	(1.97)	(1.08)	(1.24)	(1.25)	(1.17)	(0.83)	(0.89)	(0.9)
White/Asian	—	—	-1.41	-0.78	—	—	-1.45	-0.83
			(1.73)	(1.72)			(1.74)	(1.75)
Girl	—	—	3.22	3.21	—	—	3.18	3.18
			(0.8)	(0.81)			(0.8)	(0.81)
Free lunch	—	—	-13.62	-13.66	—	—	-13.68	-13.71
			(0.98)	(0.98)			(1.01)	(1.01)
White teacher	—	—	—	-0.74	—	—	—	-0.64
				(1.87)				(1.89)
Male teacher	—	—	—	1.43	—	—	—	0.97
				(3.92)				(4.28)
Teacher experience	—	—	—	0.12	—	—	—	0.13
				(0.07)				(0.07)
Master's degree	—	—	—	-2.07	—	—	—	-2.38
				(2.07)				(2.05)
School FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes

D. Third grade

Explanatory Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small class	5.59	5.72	3.28	3.01	5.84	5.56	4.4	4.2
	(1.86)	(1.16)	(1.32)	(1.35)	(1.42)	(1.03)	(1.16)	(1.18)
Regular/aide class	-0.36	-0.1	-0.39	-0.55	-0.35	0.15	0.75	0.85
	(1.89)	(1.08)	(1.32)	(1.33)	(1.13)	(0.82)	(1.04)	(1.04)
White/Asian	—	—	-0.98	-0.55	—	—	-1.33	-0.94
			(1.86)	(1.83)			(1.9)	(1.86)
Girl	—	—	3.57	3.62	—	—	3.59	3.63
			(0.89)	(0.89)			(0.89)	(0.89)
Free lunch	—	—	-11.79	-11.42	—	—	-11.9	-11.52
			(1.12)	(1.13)			(1.13)	(1.14)

White teacher	—	—	—	0.63 (2.08)	—	—	—	0.12 (2.08)
Male teacher	—	—	—	-7.11 (2.83)	—	—	—	-6.57 (2.63)
Teacher experience	—	—	—	0.06 (0.07)	—	—	—	0.06 (0.07)
School FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes

TABLE 6
EXPLORATION OF EFFECT OF ATTRITION DEPENDENT VARIABLE: AVERAGE PERCENTILE SCORE ON SAT

Actual test data			Actual and imputed test data			
Grade	Coefficient on small class dum.	Sample size		Grade	Coefficient on small class dum.	Sample size
K	5.34	5,859		K	5.34	5,859
	-1.24				-1.24	
1	5.55	4,412		1	5.04	6,100
	-1.15				-0.9	
2	3.17	3,474		2	4.09	4,555
	-1.15				-1.18	
3	4.39	3,098		3	4.34	6,116
	-1.14				-0.59	

TABLE 7
OLS AND 2SLS ESTIMATES OF EFFECT OF CLASS SIZE ON ACHIEVEMENT
DEPENDENT VARIABLE: AVERAGE PERCENTILE SCORE ON SAT

Grade	OLS	2SLS	Sample size
K	-0.63 (0.14)	-0.71 (0.14)	5,841
1	-0.82 (0.13)	-0.85 (0.16)	4,302
2	-0.42 (0.13)	-0.49 (0.16)	3,300
3	-0.36 (0.13)	-0.65 (0.17)	2,972

Figure I. Distribution of Average SAT Percentile Scores